THE TYPE OF WEIGHTED r-PATH IDEALS

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Abstract.

1. Introduction

Assumption 1.1. Throughout, let G be a (finite simple) graph with vertex set $V = V(G) = \{v_1, \dots, v_d\}$ and edge set $E = E(G)$. An edge between vertices v_i and v_j is denoted v_iv_j. Let $\mathbb{K}$ be a field, and set $R = \mathbb{K}[X_1, \dots, X_d]$. Set $\mathfrak{m} = (X_1, \dots, X_d)R$. Fix an integer $r \in \mathbb{N} = \{1, 2, \dots\}$ and set $R' = \mathbb{K}\big[\{X_{i,j} \mid i = 1, \dots, d, j = 0, \dots, r\}\big]$.

Combinatorial commutative algebra is a branch of mathematics that uses combinatorics and graph theory to understand certain algebraic constructions; it also uses algebra to understand certain objects in combinatorics and graph theory. In this paper, we explore aspects of this area via edge ideals and path ideals of graphs and weighted graphs.

Let G be a (finite simple) graph with vertex set $V = V(G) = \{v_1, \dots, v_d\}$ and edge set $E = E(G)$. Let A be a field, and consider the polynomial ring $R = A[X_1, \dots, X_d]$. The *edge ideal* of G is the ideal $I(G)$ of R that is "generated by the edges of G."

$$I(G) = (X_iX_j \mid v_iv_j \in E)R.$$

Some research in combinatorial commutative algebra uses graph-theoretic properties of G to understand algebraic properties of $I(G)$, and vice versa.

An important concept in commutative algebra is the "Cohen-Macaulay" property. The definition is somewhat technical. For now, the reader should understand that Cohen-Macaulay ideals in polynomial rings are particularly nice. If I is a Cohen-Macaulay ideal in R, the "type" of R/I roughly measures how nice the ideal is. For instance, some of the nicest Cohen-Macaulay ideals are the "Gorenstein" ideals, which end up being the Cohen-Macaulay ideals of type 1.

If G is a tree, a theorem of Villarreal [8] characterizes when $I(G)$ is Cohen-Macaulay. This characterization is purely graph-theoretical. One of the first goals of this paper is to compute the type of $R/I(G)$ for arbitrary Cohen-Macaulay trees. We accomplish this in Corollary 3.39(c). As with Villarreal's result, this computation is purely graph-theoretical. In subsequent results of this paper, we compute the type for other graph-theoretic algebra constructions the edge ideal of a weighted tree and the path ideal of a tree when they are Cohen-Macaulay. These results are in Theorems 3.37 and Corollary 3.39. These are the main results of this paper. They form the bulk of Section 3. Necessary background information is collected in Section 2.

2. Definitions

Definition 2.1. Let $D = \operatorname{depth}(R/I)$. The *type* of R/I is defined to be the dimension of the $\mathbb{K}$-vector space $\operatorname{Ext}^D_R(\mathbb{K}, R/I)$:

$$\operatorname{type}(R/I) = \dim_{\mathbb{K}}(\operatorname{Ext}^D_R(\mathbb{K}, R/I)),$$

which is a positive integer.

Fact 2.2. If I is a monomial ideal and it has an irredundant parametric decomposition $I = \bigcap_{i=1}^{t} Q_i$, then $\operatorname{type}(R/I) = t$.

3. Weighted Path Ideals and the Type of $I_r((\Sigma_r G)_\lambda)$

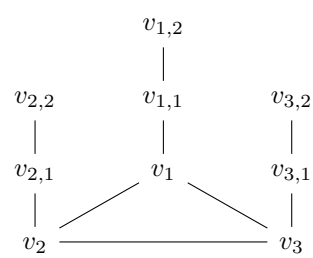
Definition 3.1. [6, Definition 1.2] A *weight function* on a graph G is a function $\omega : E \to \mathbb{N}$ that assigns a *weight* to each edge. A *weighted graph* G_ω is a graph G equipped with a weight function ω.

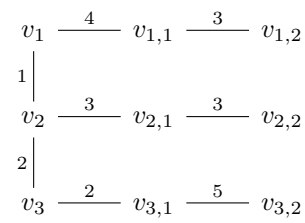
Definition 3.2. [4, Definition 3.4] The *r-path suspension* of G is the graph $\Sigma_r G$ obtained by adding a new path of length r to each vertex of G such that the vertex set

$$V(\Sigma_r G) = \{v_{i,j} \mid i = 1, \dots, d, j = 0, \dots, r\} \text{ with } v_{i,0} = v_i,\ \forall\, i = 1, \dots, d.$$

The new r-paths are called *r-whiskers*. A *weighted r-path suspension* of G_ω is a weighted graph $(\Sigma_r G)_\lambda$ with weight function $\lambda : \Sigma_r G \to \mathbb{N}$ such that the underlying graph $\Sigma_r G$ is a r-path suspension of G and $\lambda(v_i v_j) = \omega(v_i v_j)$ for all $v_i v_j \in E(G)$, i.e., $\lambda|_{E(G)} = \omega$.

Example 3.3. A 2-path suspension $\Sigma_2 C_3$ of a 3-cycle $C_3 = \big(\ v_1 \text{ —— } v_2 \text{ —— } v_3 \text{ —— } v_1\ \big)$ and a weighted 2-path suspension $(\Sigma_2 P_2)_\lambda$ of a weighted 2-path $(P_2)_\omega = \big(\ v_1 \overset{1}{\text{——}} v_2 \overset{2}{\text{——}} v_3\ \big)$ are showed in the following.





Definition 3.4. We have the following definitions:

(a) The *weighted r-path ideal* associated to G_ω is the ideal $I_r(G_\omega)$ of R that is "generated by the max-weighted paths in G of length r":

$$I_r(G_\omega) = \left(X_{i_1}^{e_{i_1}} \dots X_{i_{r+1}}^{e_{i_{r+1}}} \;\middle|\; \begin{array}{l} v_{i_1} \dots v_{i_{r+1}} \text{ is a path in } G \text{ with } e_{i_1} = \omega(v_{i_1} v_{i_2}), \\ e_{i_j} = \max\{\omega(v_{i_{j-1}} v_{i_j}), \omega(v_{i_j}, v_{i_{j+1}})\} \text{ for } 1 < j \le r \\ \text{and } e_{i_{r+1}} = \omega(v_{i_r} v_{i_{r+1}}) \end{array} \right) R.$$

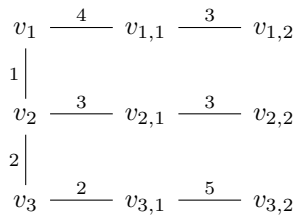
(b) Let (V', δ') be a pair such that $V' \subseteq V$ and $\delta' : V' \to \mathbb{N}$. Set $P(V', \delta') \subseteq R$ to be the ideal "generated by the elements of (V', δ')":

$$P(V', \delta') = \Big(X_i^{\delta'(v_i)} \mid v_i \in V' \Big) R.$$

Remark. Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω. Then the weighted r-path ideal $I_r\big((\Sigma_r G)_\lambda\big)$ is an ideal of R'.

Convention 3.5. Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω. Then there is a bijection between vertices of $\Sigma_r G$ and variables of $R' = \mathbb{K}[X_{i,j} \mid i = 1, \dots, d, j = 0, \dots, r]$ given by $X_{i,j} \longleftrightarrow v_{i,j}$. Based on the convention that $v_{i,0} = v_i$ for $i = 1, \dots, d$, we set $X_{i,0} = X_i$ for $i = 1, \dots, d$.

Example 3.6. Consider the following weighted graph $(\Sigma_2 P_2)_\lambda$ as in Example 3.3.



The weighted 2-path ideal associated to $(\Sigma_2 P_2)_\lambda$ is

$$I_2\big((\Sigma_2 P_2)_\lambda\big) = \big(X_{1,2}^3 X_{1,1}^4 X_1^4, X_{1,1}^4 X_1^4 X_2, X_1 X_2^3 X_{2,1}^3, X_1 X_2^2 X_3^2, X_{2,2}^3 X_{2,1}^3 X_2^3, X_{2,1}^3 X_2^3 X_3^2, X_2^2 X_3^2 X_{3,1}^2, X_{3,2}^5 X_{3,1}^5 X_3^2\big) R'.$$

Let $V' = \{v_1, v_{2,1}, v_3\} \subseteq V\big((\Sigma_2 P_2)_\lambda\big)$ and $\delta' : V' \longrightarrow \mathbb{N}$ be defined by $v_1 \longmapsto 3$, $v_{2,1} \longmapsto 2$ and $v_3 \longmapsto 4$. Then

$$P(V', \delta') = \big(X_1^3, X_{2,1}^2, X_3^4\big) R',$$

where $R' = \mathbb{K}[X_1, X_{1,1}, X_{1,2}, X_2, X_{2,1}, X_{2,2}, X_3, X_{3,1}, X_{3,2}]$.

Definition 3.7. Define a ring homomorphism p by

$$\begin{aligned} p : R' &\longrightarrow R \\ a &\longrightarrow a \ \forall\ a \in \mathbb{K}, \\ X_{ij} &\longmapsto X_i \ \forall\ i = 1, \dots, d,\ j = 0, \dots, r. \end{aligned}$$

Let $I \subseteq R'$ be a monomial ideal and set

$$IR = p(I)R = \big(X_{i_1}^{a_1} \dots X_{i_n}^{a_n} \in R \mid \exists\ X_{i_1,j_1}^{a_1} \cdots X_{i_n,j_n}^{a_n} \in [\![I]\!]\big) R.$$

In words, IR is the monomial ideal of R obtained from I by setting $X_{i,j} = X_i$ for all i, j. It is straightforward to show that if $f_1, \dots, f_m$ is a monomial generating sequence for I, then $p(f_1), \dots, p(f_m)$ is a monomial generating sequence for IR.

Example 3.8. Consider the 2-path ideal $I_2(\Sigma_2 P_2)$ from Example 3.6. Then

$$\begin{aligned} I_2\big((\Sigma_2 P_2)_\lambda\big) R &= \big(X_1^3 X_1^4 X_1^4, X_1^4 X_1^4 X_2, X_1 X_2^3 X_2^3, X_1 X_2^2 X_3^2, X_2^3 X_2^3 X_3^2, \\ &\qquad X_2^3 X_2^3 X_3^2, X_2^2 X_3^2 X_3^2, X_3^5 X_3^5 X_3^2\big) R \\ &= (X_1^{11}, X_1^8 X_2, X_1 X_2^6, X_1 X_2^2 X_3^2, X_2^9, X_2^6 X_3^2, X_2^2 X_3^4, X_3^{12}) R. \end{aligned}$$

Definition 3.9. Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω. Define

$$\mathfrak{m}^{[\underline{a}(\lambda)]} = \big(X_1^{a_1}, \dots, X_d^{a_d}\big) R,$$

where for $i = 1, \dots, d$: $a_i = \sum_{k=0}^{r} e_{i,k}$ with

$$e_{i,k} = \begin{cases} \lambda(v_i v_{i,1}) & \text{if } k = 0, \\ \max\{\lambda(v_{i,k-1} v_{i,k}), \lambda(v_{i,k} v_{i,k+1})\} & \text{if } k = 1, \dots, r-1, \\ \lambda(v_{i,r-1} v_{i,r}) & \text{if } k = r. \end{cases}$$

In words, $\mathfrak{m}^{\underline{a}(\lambda)}$ is the monomial ideal of R obtained from the monomial ideal $(g_1, \dots, g_d)R'$ by setting $\mathfrak{m}^{\underline{a}(\lambda)} = \big(p(g_1), \dots, p(g_d)\big) R$, where g_i is the corresponding generator in $I_r\big((\Sigma_r G)_\lambda\big)$ of the r-whisker $v_i v_{i,1} \dots v_{i,r}$ from $(\Sigma_r G)_\lambda$ for $i = 1, \dots, d$.

Example 3.10. In Example 3.6, $\mathfrak{m}^{[\underline{a}(\lambda)]} = \big(X_1^{a_1}, X_2^{a_2}, X_3^{a_3}\big)R$ with

$$a_1 = \sum_{k=0}^{2} e_{1,k} = \lambda(v_1v_{1,1}) + \max\{\lambda(v_1v_{1,1}), \lambda(v_{1,1}v_{1,2})\} + \lambda(v_{1,1}v_{1,2}) = 4+4+3 = 11,$$
$$a_2 = \sum_{k=0}^{2} e_{2,k} = \lambda(v_2v_{2,1}) + \max\{\lambda(v_2v_{2,1}), \lambda(v_{2,1}v_{2,2})\} + \lambda(v_{2,1}v_{2,2}) = 3+3+3 = 9,$$
$$a_3 = \sum_{k=0}^{2} e_{3,k} = \lambda(v_3v_{3,1}) + \max\{\lambda(v_3v_{3,1}), \lambda(v_{3,1}v_{3,2})\} + \lambda(v_{3,1}v_{3,2}) = 2+5+5 = 12.$$

Fact 3.11. Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω such that $\lambda(v_iv_j) \le \lambda(v_i, v_{i,1})$ and $\lambda(v_iv_j) \le \lambda(v_j, v_{j,1})$ for all edges $v_iv_j \in E$. Then $I_r\big((\Sigma_r G)_\lambda\big)$ is the polarization of $I_r\big((\Sigma_r G)_\lambda\big)R$ by e.g., [4, Proposition 3.7]. So by [1], the list $X_i - X_{i,k}, 1 \le i \le d, 1 \le k \le r$ is a maximal homogeneous regular sequence for $\frac{R'}{I_r((\Sigma_r G)_\lambda)}$ and

$$\frac{R}{I_r((\Sigma_r G)_\lambda)R} \cong \frac{R'}{I_r((\Sigma_r G)_\lambda) + (X_i - X_{i,k} \mid 1 \le i \le d, 1 \le k \le r)R'}.$$

Furthermore, it is straightforward to show that

$$I_r\big((\Sigma_r G)_\lambda\big)R = I_r\big((\Sigma_{r-1} G)_{\lambda'}\big)R + \mathfrak{m}^{[\underline{a}(\lambda)]}, \text{ where } \lambda' = \lambda|_{\Sigma_{r-1}G}.$$

Because of the following fact, the main result of this section gives a formula to compute the type$\big(R/I_r(G_\omega)\big)$ for all trees such that $R/I_r(G_\omega)$ is Cohen-Macaulay.

Fact 3.12. [4, Proposition 3.7 and Theorem 3.11] Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω such that $\lambda(v_iv_j) \le \min\{\lambda(v_i, v_{i,1}), \lambda(v_j, v_{j,1})\}$ for all edges $v_iv_j \in E$.

(a) $R'/I_r\big((\Sigma_r G)_\lambda\big)$ is Cohen-Macaulay.
(b) If $\Gamma_{\lambda'}$ is a weighted tree and $R/I_r(\Gamma_{\lambda'})$ is Cohen-Macaulay, then there exists a weighted tree $H_{\omega'}$ such that $(\Sigma_r H)_{\lambda''}$ is obtained by pruning a sequence of r-pathless leaves from $\Gamma_{\lambda'}$ with $\lambda'' = \lambda'|_{\Sigma_r H}$ and the weight function λ' satisfies the above condition.

Definition 3.13. Let $(\Sigma_{r-1} G)_\lambda$ be a weighted $(r-1)$-path suspension of G_ω. We define $q : V\big((\Sigma_{r-1} G)_\lambda\big) \to V(G)$ as $q(v_{i,j}) = v_i$. Let $\mathfrak{P} := (V'', \delta'')$ be such that $V'' \subseteq V\big((\Sigma_{r-1} G)_\lambda\big)$ and $\delta'' : V'' \to \mathbb{N}$. Then

$$q(V'') = \{v_i \mid \exists\, v_{i,j} \in V''\}.$$

Set

$$\mathrm{WCA}_i(\mathfrak{P}) = \{v_{i,j} \in V'' \mid \delta''(v_{i,j}) \le \lambda(v_{i,j}v) \text{ for some edge } v_{i,j}v \text{ in } (\Sigma_{r-1} G)_\lambda\},\ \forall\, i = 1, \dots, d,$$

$$h_{i,k} = \max\big\{\lambda(v_{i,k}v) \mid v_{i,k}v \in E\big((\Sigma_{r-1} G)_\lambda\big)\big\},\ \forall\, i = 1, \dots, d,\ k = 0, \dots, r-1.$$

Define

$$\gamma_{(V'',\delta'')} : q(V'') \longrightarrow \mathbb{N} \sqcup \{\infty\}$$
$$v_i \longmapsto \begin{cases} \min\Big\{\delta''(v_{i,j}) + \sum_{k=0}^{j-1} h_{i,k} \;\Big|\; v_{i,j} \in \mathrm{WCA}_i(\mathfrak{P})\Big\} & \text{if } \mathrm{WCA}_i(\mathfrak{P}) \neq \emptyset, \\ \infty & \text{otherwise.} \end{cases}$$

Proposition 3.14. Let $(\Sigma_{r-1}G)_\lambda$ be a weighted $(r-1)$-path suspension of G_ω. Let $\mathfrak{P} := (V'', \delta'')$ be such that $V'' \subseteq V\big((\Sigma_{r-1}G)_\lambda\big)$ and $\delta'' : V'' \to \mathbb{N}$. If $\mathrm{WCA}_i(\mathfrak{P}) \neq \emptyset$ for some $i \in \{1, \dots, d\}$, then for any $v_i \in q(V'')$,

$$\gamma_{(V'',\delta'')}(v_i) = \delta''(v_{i,j_0}) + \sum_{k=0}^{j_0-1} h_{i,k}, \text{ where } j_0 := \min\{j \mid v_{i,j} \in \mathrm{WCA}_i(\mathfrak{P})\}.$$

Proof. Suppose $\delta''(v_{i,j}) + \sum_{k=0}^{j-1} h_{i,k} < \delta''(v_{i,j_0}) + \sum_{k=0}^{j_0-1} h_{i,k}$ for some $v_{i,j} \in \mathrm{WCA}_i(\mathfrak{P})$. Then by the definition of j_0, we have $j > j_0 \geq 0$ and so $\delta''(v_{i,j}) + \sum_{k=j_0}^{j-1} h_{i,k} < \delta''(v_{i,j_0})$. Thus,

$$h_{i,j_0} < \delta''(v_{i,j}) + h_{i,j_0} \leq \delta''(v_{i,j}) + \sum_{k=j_0}^{j-1} h_{i,k} < \delta''(v_{i,j_0}), \text{ i.e., } h_{i,j_0} < \delta''(v_{i,j_0}),$$

contradicted by that $v_{i,j_0} \in \mathrm{WCA}_i(\mathfrak{P})$. □

The following theorem is a key for decomposing $I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R$ with $\lambda' = \lambda|_{\Sigma_{r-1}G}$ and hence $I_r\big((\Sigma_r G)_\lambda\big)R$. The proof is somewhat technical. The reader may wish to follow the argument with the succeeding example.

Theorem 3.15. *Let $(\Sigma_{r-1}G)_\lambda$ be a weighted $(r-1)$-path suspension of G_ω such that $\lambda(v_iv_j) \leq \lambda(v_i, v_{i,1})$ and $\lambda(v_iv_j) \leq \lambda(v_jv_{j,1})$ for all edges $v_iv_j \in E$. Let $\mathfrak{P} := (V'', \delta'')$ be such that $V'' \subseteq V\big((\Sigma_{r-1}G)_\lambda\big)$ and $\delta'' : V'' \to \mathbb{N}$. Then $I_r\big((\Sigma_{r-1}G)_\lambda\big)R \subseteq P\big(q(V''), \gamma_{(V'',\delta'')}\big)$ if and only if (V'', δ'') is a weighted r-path vertex cover of $(\Sigma_{r-1}G)_\lambda$.*

Proof. "⇒". Assume $I_r\big((\Sigma_{r-1}G)_\lambda\big)R \subseteq P\big(q(V''), \gamma_{(V'',\delta'')}\big)$. Let $P_r := v_{p_1,q_1} \dots v_{p_{r+1},q_{r+1}}$ be an r-path in $(\Sigma_{r-1}G)_\lambda$. Set

$$e_{p_k,q_k} = \begin{cases} \lambda(v_{p_1,q_1}v_{p_2,q_2}) & \text{if } k = 1, \\ \max\{\lambda(v_{p_{k-1},q_{k-1}}v_{p_k,q_k}), \lambda(v_{p_k,q_k}v_{p_{k+1},q_{k+1}})\} & \text{if } k = 2, \dots, r, \\ \lambda(v_{p_r,q_r}v_{p_{r+1},q_{r+1}}) & \text{if } k = r+1. \end{cases}$$

Then $X_{p_1}^{e_{p_1,q_1}} \dots X_{p_{r+1}}^{e_{p_{r+1},q_{r+1}}} \in [\![I_r((\Sigma_{r-1}G)_\lambda)R]\!] \subseteq [\![P(q(V''), \gamma_{(V'',\delta'')})]\!]$. So

$$X_{i_0}^{\gamma_{(V'',\delta'')}(v_{i_0})} \mid X_{p_1}^{e_{p_1,q_1}} \dots X_{p_{r+1}}^{e_{p_{r+1},q_{r+1}}} \text{ for some } v_{i_0} \in q(V'').$$

Hence $v_{i_0} = v_{p_l}$ for some $l \in \{1, \dots, r+1\}$, $\gamma_{(V'',\delta'')(v_{i_0})} < \infty$ and so

$$\min_{v_{i_0,j} \in \mathrm{WCA}_{i_0}(\mathfrak{P})} \left\{ \delta''(v_{i_0,j}) + \sum_{k=0}^{j-1} h_{i_0,k} \right\} = \gamma_{(V'',\delta'')}(v_{i_0}) \leq \sum_{k=0}^{r+1} \mathbb{1}_k \cdot e_{p_k,q_k}, \text{ where } \mathbb{1}_k = \begin{cases} 1 & \text{if } p_k = i_0, \\ 0 & \text{otherwise.} \end{cases}$$

So $v_{p_l} = v_{i_0} \in q(V'')$. Since P_r is an r-path in $\Sigma_{r-1}G$, P_r is of the following form.

$$\begin{array}{ccccccc}
v_{p_{1+q_1},0} & & & & & & \\
\| & & & & & & \\
v_{p_1,0} & \text{———} & v_{p_1,1} & \text{———} & \cdots & \text{———} & v_{p_1,q_1} \\
| & & & & & & \\
\vdots & & & & & & \\
| & & & & & & \\
v_{p_{r+1},0} & \text{———} & v_{p_{r+1},1} & \text{———} & \cdots & \text{———} & v_{p_{r+1},q_{r+1}} \\
\| & & & & & & \\
v_{p_{1+r-q_{r+1}},0} & & & & & &
\end{array}$$

where q_1 or q_{r+1} may be 0. Let $M_0 := \max_{1\le k\le r+1}\{q_k \mid i_0 = p_k\}$. Then

$$M_0 = \begin{cases} q_1 & \text{if } i_0 = p_1, \\ q_{r+1} & \text{if } i_0 = p_{r+1}. \end{cases}$$

Since $\gamma_{(V'',\delta'')}(v_i) < \infty$, we have $\mathrm{WCA}_{i_0}(\mathfrak{P}) \neq \emptyset$. Set $j_0 := \min\{j \mid v_{i_0,j} \in \mathrm{WCA}_{i_0}(\mathfrak{P})\}$. Then by Proposition 3.14,

$$\delta''(v_{i_0,j_0}) + \sum_{k=0}^{j_0-1} h_{i_0,k} = \min_{v_{i_0,j}\in V''}\left\{\delta''(v_{i_0,j}) + \sum_{k=0}^{j-1} h_{i_0,k}\right\} \le \sum_{k=0}^{r+1} \mathbb{1}_k \cdot e_{p_k,q_k} = \sum_{k=0}^{M_0} e_{i_0,k}. \tag{3.1}$$

Suppose $j_0 > M_0$. Then since $e_{i_0,k} \le h_{i_0,k}$ for $k = 0,\dots,M_0$, by (3.1), we have

$$\delta''(v_{i_0,j_0}) + \sum_{k=0}^{M_0} e_{i_0,k} \le \delta''(v_{i_0,j_0}) + \sum_{k=0}^{M_0} h_{i_0,k} \le \delta''(v_{i_0,j_0}) + \sum_{k=0}^{j_0-1} h_{i_0,k} \le \sum_{k=0}^{M_0} e_{i_0,k}, \text{ i.e., } \delta''(v_{i_0,j_0}) \le 0,$$

contradicted by that $\delta''(v_{i_0,j_0}) \ge 1$. So $j_0 \le M_0$. Also, we have there must exist a sub-path of P_r of the form

$$v_{i_0,0} \text{ ——— } v_{i_0,1} \text{ ——— } \cdots \text{ ——— } v_{i_0,M_0},$$

so there exists a vertex in this path of the form $v_{i_0,j_0} = v_{p_k,q_k}$ for some k in $\{1,\dots,r+1\}$. So $v_{p_k,q_k} = v_{i_0,j_0} \in \mathrm{WCA}_i(\mathfrak{P}) \subseteq V''$ and $v_{i_0,j_0} \in V(P_r)$.

(a) Assume $0 = j_0 < M_0$. Since $\lambda(v_iv_j) \le \min\{\lambda(v_i,v_{i,1}),\lambda(v_j,v_{j,1})\}$ for all edges $v_iv_j \in E$ and $M_0 \ge 1$, we have $e_{i_0,0} = \lambda(v_{i_0,0}v_{i_0,1}) = h_{i_0,0}$. Since $v_{i_0,j_0} \in \mathrm{WCA}_{i_0}(\mathfrak{P})$, we have $\delta''(v_{i_0,0}) \le h_{i_0,0} = e_{i_0,0}$.

(b) Assume $0 < j_0 < M_0$. Since $v_{i_0,j_0} \in \mathrm{WCA}_{i_0}(\mathfrak{P})$, we have

$$\delta''(v_{i_0,j_0}) \le \max\{\lambda(v_{i_0,j_0-1}v_{i_0,j_0}),\lambda(v_{i_0,j_0}v_{i_0,j_0+1})\} = e_{i_0,j_0}.$$

(c) Assume $j_0 = M_0$. Since $e_{i_0,k} \le h_{i_0,k}$ for $k = 0,\dots,j_0-1$, by (3.1), we have

$$\delta''(v_{i_0,j_0}) + \sum_{k=0}^{j_0-1} e_{i_0,k} \le \delta''(v_{i_0,j_0}) + \sum_{k=0}^{j_0-1} h_{i_0,k} \le \sum_{k=0}^{M_0} e_{i_0,k} = \sum_{k=0}^{j_0} e_{i_0,k}, \text{ i.e., } \delta''(v_{i_0,j_0}) \le e_{i_0,j_0}.$$

So we have $v_{i_0,j_0} \in V'' \cap V(P_r)$ such that $\delta''(v_{i_0,j_0}) \le e_{i_0,j_0}$ by (a), (b) and (c). Thus, (V'',δ'') is a weighted r-path vertex cover of $\Sigma_{r-1}G$.

"$\Leftarrow$". Assume (V'', δ'') is a weighted r-path vertex cover of $(\Sigma_{r-1}G)_\lambda$. We need to show every monomial generator of $I_r\big((\Sigma_{r-1}G)_\lambda\big)R$ is in $P\big(q(V''), \gamma_{(V'',\delta'')}\big)$. Let $\underline{X}^{\underline{b}} := X_{i_1}^{e_{i_1,j_1}} \dots X_{i_{r+1}}^{e_{i_{r+1},j_{r+1}}}$ be such a generator corresponding to an r-path $P_r := v_{i_1,j_1} \dots v_{i_{r+1},j_{r+1}}$ in $(\Sigma_{r-1}G)_\lambda$. We need to show $\underline{X}^{\underline{b}} \in P\big(q(V''), \gamma_{(V'',\delta'')}\big)$. Note that $X_{i_1,j_1}^{e_{i_1,j_1}} \dots X_{i_{r+1},j_{r+1}}^{e_{i_{r+1},j_{r+1}}}$ is of the following form.

$$\begin{array}{ccccccc}
X_{i_{1+j_1},0}^{e_{i_{1+j_1},0}} & & & & & & \\
\| & & & & & & \\
X_{i_1,0}^{e_{i_1,0}} & \text{———} & X_{i_1,1}^{e_{i_1,1}} & \text{———} & \cdots & \text{———} & X_{i_1,j_1}^{e_{i_1,j_1}} \\
| & & & & & & \\
\vdots & & & & & & \\
| & & & & & & \\
X_{i_{r+1},0}^{e_{i_{r+1},0}} & \text{———} & X_{i_{r+1},1}^{e_{i_{r+1},1}} & \text{———} & \cdots & \text{———} & X_{i_{r+1},j_{r+1}}^{e_{i_{r+1},j_{r+1}}} \\
\| & & & & & & \\
X_{i_{1+r-j_{r+1}},0}^{e_{i_{1+r-j_{r+1}},0}} & & & & & &
\end{array}$$

where j_1 or j_{r+1} may be 0. Since P_r is an r-path in $(\Sigma_{r-1}G)_\lambda$ and (V'', δ'') is a weighted r-path vertex cover of $(\Sigma_{r-1}G)_\lambda$, we have there exists some $l \in \{1, \dots, r+1\}$ such that $v_{i_l,j_l} \in V''$ and $\delta''(v_{i_l,j_l}) \le e_{i_l,j_l}$. So $v_{i_l,j_l} \in \mathrm{WCA}_{i_l}(\mathfrak{P})$ and then

$$\gamma_{(V'',\delta'')}(v_{i_l}) = \min_{v_{i_l,t} \in \mathrm{WCA}_{i_l}(\mathfrak{P})} \left\{ \delta''(v_{i_l,t}) + \sum_{k=0}^{t-1} h_{i_l,k} \right\} \le \delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} h_{i_l,k}.$$

Let $M_0 := \max_{1 \le k \le r+1}\{j_k \mid i_l = i_k\}$. Then $j_l \le M_0$. So

$$\delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} e_{i_l,k} \le e_{i_l,j_l} + \sum_{k=0}^{j_l-1} e_{i_l,k} = \sum_{k=0}^{j_l} e_{i_l,k} \le \sum_{k=0}^{M_0} e_{i_l.k}, \text{ i.e., } \delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} e_{i_l,k} \le \sum_{k=0}^{M_0} e_{i_l,k}.$$

(a) Assume $j_l = 0$. Then

$$\gamma_{(V'',\delta'')}(v_{i_l}) \le \delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} h_{i_l,k} = \delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} e_{i_l,k} \le \sum_{k=0}^{M_0} e_{i_l,k} = \sum_{k=0}^{r+1} \mathbb{1}_{l,k} \cdot e_{i_k,j_k} = b_{i_l},$$

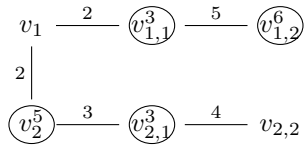
where $\mathbb{1}_{l,k} = \begin{cases} 1 & \text{if } j_k = j_l, \\ 0 & \text{otherwise,} \end{cases} \quad \forall\, k = 1, \dots, r+1.$

(b) Assume $j_l > 0$. Then $M_0 \ge 1$. Since $\lambda(v_i v_j) \le \min\{\lambda(v_i, v_{i,1}), \lambda(v_j, v_{j,1})\}$ for all edges $v_i v_j \in E$, we have $e_{i_0,0} = \lambda(v_{i_0,0} v_{i_0,1}) = h_{i_0,0}$. Also, since $e_{i_0,k} = h_{i_0,k}$ for $k = 1, \dots, j_l - 1$, we have

$$\gamma_{(V'',\delta'')}(v_{i_l}) \le \delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} h_{i_l,k} = \delta''(v_{i_l,j_l}) + \sum_{k=0}^{j_l-1} e_{i_l,k} \le \sum_{k=0}^{M_0} e_{i_l,k} = \sum_{k=0}^{r+1} \mathbb{1}_{l,k} \cdot e_{i_k,j_k} = b_{i_l}.$$

So $X_{i_l}^{\gamma_{(V'',\delta'')}(v_{i_l})} \mid \underline{X}^{\underline{b}}$ by (a) and (b). Thus, $\underline{X}^{\underline{b}} \in P\big(q(V''), \gamma_{(V'',\delta'')}\big)$. $\square$

Example 3.16. A weighted 2-path suspension $(\Sigma_2 P_1)_\lambda$ of $(P_1)_\omega = \big(\ v_1 \overset{2}{\text{———}} v_2\ \big)$ with a weighted 3-path vertex cover $\mathfrak{P} := (V'', \delta'')$ of $(\Sigma_2 P_1)_\lambda$ is given in the following sketch.



Since $I_3\big((\Sigma_2 P_1)_\lambda\big) = \big(X^5_{1,2}X^5_{1,1}X^2_1X^2_2, X^2_{1,1}X^2_1X^3_2X^3_{2,1}, X^2_1X^3_2X^4_{2,1}X^4_{2,2}\big)R'$, we have

$$I_3\big((\Sigma_2 P_1)_\lambda\big)R = \big(X^{12}_1X^2_2, X^4_1X^6_2, X^2_1X^{11}_2\big)R.$$

Note that $q(V'') = \{v_1, v_2\}$. Since $\delta''(v_{1,1}) = 3 < 5 = \lambda(v_{1,1}v_{1,2})$ and $\delta''(v_{1,2}) = 6 > 5 = \lambda(v_{1,1}v_{1,2})$, we have $\mathrm{WCA}_1(\mathfrak{P}) = \{v_{1,1}\}$. Similarly, $\mathrm{WCA}_2(\mathfrak{P}) = \{v_{2,1}\}$. So

$$\gamma_{(V'',\delta'')}(v_1) = \delta''(v_{1,1}) + \sum_{k=0}^{1-1} h_{1,k} = \delta''(v_{1,1}) + \max\{\lambda(v_1v_2), \lambda(v_1v_{1,1})\} = 3 + \max\{2,2\} = 5,$$

$$\gamma_{(V'',\delta'')}(v_2) = \delta''(v_{2,1}) + \sum_{k=0}^{1-1} h_{2,k} = \delta''(v_{2,1}) + \max\{\lambda(v_1v_2), \lambda(v_2v_{2,1})\} = 3 + \max\{2,3\} = 6.$$

Then

$$P\big(V'', \gamma_{(q(V''),\delta'')}\big) = \big(X^5_1, X^6_2\big)R \supseteq \big(X^{12}_1X^2_2, X^4_1X^6_2, X^2_1X^{11}_2\big)R = I_3\big((\Sigma_2 P_1)_\lambda\big)R.$$

Proposition 3.17. Let $(\Sigma_{r-1}G)_\lambda$ be a weighted $(r-1)$-path suspension of G_ω such that $\lambda(v_iv_j) \le \lambda(v_i, v_{i,1})$ and $\lambda(v_iv_j) \le \lambda(v_j, v_{j,1})$ for all $v_iv_j \in E$. The monomial ideal $I_r\big((\Sigma_{r-1}G)_\lambda\big)R$ can be written as a finite intersection of m-irreducible ideals of the form $P\big(q(V'') := \{v_{i_1}, \dots, v_{i_t}\}, \gamma_{(V'',\delta'')}\big)$ with $V'' \subseteq V(\Sigma_{r-1}G)$ and $\delta'' : V'' \to \mathbb{N}$.

Proof. [5, Theorem 7.5.1] gives a decomposition of $I_r\big((\Sigma_{r-1}G)_\lambda\big)R$. Let $\big(X^{\beta_{b_1}}_{b_1}, \dots, X^{\beta_{b_s}}_{b_s}\big)R$ occur in that decomposition. Without loss of generality, assume $b_1, \dots, b_s \in \mathbb{N}$ are distinct. Let $k \in \{1, \dots, s\}$. Then by [5, Theorem 7.5.1], there exists a generator $p(X^{e_{i_1,j_1}}_{i_1,j_1} \dots X^{e_{i_{r+1},j_{r+1}}}_{i_{r+1},j_{r+1}})$ of $I_r\big((\Sigma_{r-1}G)_\lambda\big)R$ with $v_{i_1,j_1} \dots v_{i_{r+1},j_{r+1}}$ an r-path in $(\Sigma_{r-1}G)_\lambda$ such that

$$\beta_{b_k} = \begin{cases} e_{b_k,0} & \text{if } M_k = 0, \\ \lambda(v_{b_k,M_k}v_{b_k,M_k-1}) + \sum_{l=0}^{M_k-1} h_{b_k,l} & \text{if } M_k \ge 1, \end{cases}$$

where $M_k := \max_{1\le n\le r+1}\{j_n \mid b_k = i_n\} \le r-1$ and

$$e_{i_m,j_m} = \begin{cases} \lambda(v_{i_1,j_1}v_{i_2,j_2}) & \text{if } m = 1, \\ \max\{\lambda(v_{i_{m-1},j_{m-1}}v_{i_m,j_m}), \lambda(v_{i_m,j_m}v_{i_{m+1},j_{m+1}})\} & \text{if } m = 2, \dots, r, \\ \lambda(v_{i_r,j_r}v_{i_{r+1},j_{r+1}}) & \text{if } m = r+1. \end{cases}$$

Repeat the process for each $k \in \{1, \dots, s\}$ and set $V'' = \{v_{b_1,M_1}, \dots, v_{b_s,M_s}\}$. Then $q(V'') = \{v_{b_1}, \dots, v_{b_s}\}$. Define

$$\begin{aligned} \delta'' : V'' &\longrightarrow \mathbb{N} \\ v_{b_k,M_k} &\longmapsto \begin{cases} \lambda(v_{b_k,M_k}v_{b_k,M_k-1}) & \text{if } M_k \ge 1, \\ \beta_{b_k} (= e_{b_k,0}) & \text{if } M_k = 0, \end{cases} \quad \forall\ k = 1, \dots, s. \end{aligned}$$

Claim. $(X^{\beta_{b_1}}_{b_1}, \dots, X^{\beta_{b_s}}_{b_s})R = P\big(q(V''), \gamma_{(V'',\delta'')}\big)$. It is enough to show that $\gamma_{(V'',\delta'')}(v_{b_k}) = \beta_{b_k}$ for $k = 1, \dots, s$. Let $\mathfrak{P} := (V'', \delta'')$. By definition, we have $\mathrm{WCA}_k(\mathfrak{P}) = \{v_{b_k,M_k}\}$ for $k = 1, \dots, s$.

(a) Assume $M_k \geq 1$. Then

$$\gamma_{(V'',\delta'')}(v_{b_k}) = \delta''(v_{b_k,M_k}) + \sum_{l=0}^{M_k-1} h_{b_k,l} = \lambda(v_{b_k,M_k}v_{b_k,M_k-1}) + \sum_{l=0}^{M_k-1} h_{b_k,l} = \beta_{b_k}.$$

(b) Assume $M_k = 0$. Then

$$\gamma_{(V'',\delta'')}(v_{b_k}) = \delta''(v_{b_k,0}) + \sum_{l=0}^{0-1} h_{b_k,l} = \delta''(v_{b_k,0}) = \beta_{b_k}. \qquad \square$$

Example 3.18. Consider the following graph $(\Sigma_2 P_1)_\lambda$ as in Example 3.16.

$$\begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\Big| & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} \end{array}$$

By Example 3.16, $I_3\big((\Sigma_2 P_1)\big)R = (X_1^{12}X_2^2, X_1^4X_2^6, X_1^2X_2^{11})R$. By [5, Theorem 7.5.1],

$$\begin{aligned} I_3\big((\Sigma_2 P_1)\big)R &= \big(X_1^{12}, X_1^4, X_1^2\big)R \cap \big(X_1^{12}, X_1^4, X_2^{11}\big)R \cap \big(X_1^{12}, X_2^6, X_1^2\big)R \cap \big(X_1^{12}, X_2^6, X_2^{11}\big)R \\ &\quad \cap \big(X_2^2, X_1^4, X_1^2\big)R \cap \big(X_2^2, X_1^4, X_2^{11}\big)R \cap \big(X_2^2, X_2^6, X_1^2\big)R \cap \big(X_2^2, X_2^6, X_2^{11}\big)R \\ &= \big(X_1^2\big)R \cap \big(X_1^4, X_2^{11}\big)R \cap \big(X_1^2, X_2^6\big)R \cap \big(X_1^{12}, X_2^6\big)R \\ &\quad \cap \big(X_1^2, X_2^2\big)R \cap \big(X_1^4, X_2^2\big)R \cap \big(X_1^2, X_2^2\big)R \cap \big(X_2^2\big)R. \end{aligned}$$

Consider the m-irreducible ideal $(X_1^4, X_2^{11})R$. Then $b_1 = 1, b_2 = 2$, $\beta_{b_1} = \beta_1 = 4$ and $\beta_{b_2} = \beta_2 = 11$. Consider the generator $X_{1,1}^{e_{1,1}}X_{1,0}^{e_{1,0}}X_{2,0}^{e_{2,0}}X_{2,1}^{e_{2,1}} := X_{11}^2X_1^2X_2^3X_{2,1}^3$ of $I_3\big((\Sigma_2 P_1)_\lambda\big)$. Then $M_1 := 1$ and $\beta_1 = \lambda(v_{1,1}v_{1,0}) + h_{1,0} = 2+2 = 4$. Consider the generator $X_{1,0}^{e_{1,0}}X_{2,0}^{e_{2,0}}X_{2,1}^{e_{2,1}}X_{2,2}^{e_{2,2}} := X_1^2X_2^3X_{2,1}^4X_{2,2}^4$ of $I_3\big((\Sigma_2 P_1)_\lambda\big)$. Then $M_2 := 2$ and $\beta_2 = \lambda(v_{2,2}v_{2,1}) + h_{2,0} + h_{2,1} = 4+3+4 = 11$. Let $V'' = \{v_{1,1}, v_{2,2}\}$ and $\delta'' : V'' \longrightarrow \mathbb{N}$ given by $v_1 \longmapsto \lambda(v_{1,1}v_{1,0}) = 2$ and $v_{2,2} \longmapsto \lambda(v_{2,2}v_{2,1}) = 4$. So $q(V'') = \{v_1, v_2\}$, $\gamma_{(V'',\delta'')}(v_1) = \delta''(v_{1,1}) + h_{1,0} = 2+2$ and $\gamma_{(V'',\delta'')}(v_2) = \delta''(v_{2,2}) + h_{2,0} + h_{2,1} = 4+3+4 = 11$. Hence

$$P\big(q(V''), \gamma(V'', \delta'')\big) = P\big(\{v_1^4, v_2^{11}\}\big) = \big(X_1^4, X_2^{11}\big)R.$$

The next result gives our first decomposition needed for computing type$\big(R'/I_r((\Sigma_r G)_\lambda)\big)$.

Theorem 3.19. *Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω such that $\lambda(v_iv_j) \leq \lambda(v_iv_{i,1})$ and $\lambda(v_iv_j) \leq \lambda(v_jv_{j,1})$ for all edges $v_iv_j \in E$. One has*

$$I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R = \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big), \text{ where } \lambda' = \lambda|_{\Sigma_{r-1}G},$$

and

$$I_r\big((\Sigma_r G)_\lambda\big)R = \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big) + \mathfrak{m}^{[\underline{a}(\lambda)]}.$$

Proof. Since $I_r\big((\Sigma_r G)_\lambda\big)R = I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R + \mathfrak{m}^{[\underline{a}(\lambda)]}$ by Fact 3.11, we have by [5, Theorem 7.5.3], it is enough to show that

$$I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R = \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big).$$

By Proposition 3.17, the monomial ideal $I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R$ can be written as a finite intersection of m-irreducible ideals of the form $P\big(q(V'') := \{v_{i_1}, \dots, v_{i_t}\}, \gamma_{(V'',\delta'')}\big)$ with $V'' \subseteq V(\Sigma_{r-1}G)$ and $\delta'' : V'' \to \mathbb{N}$. Then by Theorem 3.15,

$$\begin{aligned} I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R &\subseteq \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big) \\ &\subseteq \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'} \text{ in decomp. of } I_r((\Sigma_{r-1}G)_{\lambda'})R} P\big(q(V''), \gamma_{(V'',\delta'')}\big) \\ &= I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R. \end{aligned}$$

So

$$I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R = \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big). \qquad \square$$

Example 3.20. Consider the following weighted 3-path suspension $(\Sigma_3 P_1)_\lambda$ of the weighted 1-path $(P_1)_\omega = \big(\ v_1 \overset{2}{\text{———}} v_2\ \big)$.

$$\begin{array}{ccccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2} & \overset{2}{\text{———}} & v_{1,3} \\ {\scriptstyle 2}\Big| & & & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} & \overset{2}{\text{———}} & v_{2,3} \end{array}$$

Let $\lambda' = \lambda|_{\Sigma_2 P_1}$. Since $I_3\big((\Sigma_2 P_1)_{\lambda'}\big) = (X_{1,2}^5 X_{1,1}^5 X_1^2 X_2^2, X_{1,1}^2 X_1^2 X_2^3 X_{2,1}^3, X_1^2 X_2^3 X_{2,1}^4 X_{2,2}^4)$, by Theorem 3.19, we have two infinite intersection:

$$I_3\big((\Sigma_2 P_1)_{\lambda'}\big)R = \big(X_1^{12} X_2^2, X_1^4 X_2^6, X_1^2 X_2^{11}\big)R = \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_2 P_1)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big),$$

and

$$\begin{aligned} I_3\big((\Sigma_3 P_1)_{\lambda'}\big)R &= \big(X_1^{12} X_2^2, X_1^4 X_2^6, X_1^2 X_2^{11}\big)R + (X_1^{14}, X_2^{13})R \\ &= \bigcap_{(V'',\delta'') \text{ w. } r\text{-path v. cover of } (\Sigma_2 P_1)_{\lambda'}} P\big(q(V''), \gamma_{(V'',\delta'')}\big). \end{aligned}$$

The next result is key for our second decomposition result, Corollary 3.23.

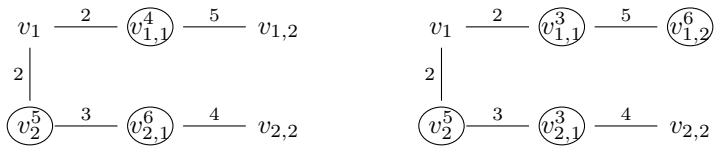
Lemma 3.21. Let $\mathfrak{p} := (V_1'', \delta_1''), \mathfrak{P} := (V_2'', \delta_2'')$ be such that $V_1'', V_2'' \subseteq V\big((\Sigma_{r-1}G)_\lambda\big)$ and $\delta_1'', \delta_2'' : V'' \to \mathbb{N}$. If $(V_1'', \delta_1'') \le (V_2'', \delta_2'')$, then $P\big(q(V_1''), \gamma_{(V_1'',\delta_1'')}\big) \subseteq P\big(q(V_2''), \gamma_{(V_2'',\delta_2'')}\big)$.

Proof. Let $X_i^{\gamma_{(V_1'',\delta_1'')}(v_i)}$ be a generator of $P\big(q(V_1''), \gamma_{(V_1'',\delta_1'')}\big)$. Then V_1''"$\subseteq V_2''$" implies we have $X_i^{\gamma_{(V_2'',\delta_2'')}(v_i)} \in P\big(q(V_2''), \gamma_{(V_2'',\delta_2'')}\big)$ and

$$\begin{aligned} \gamma_{(V_1'',\delta_1'')}(v_i) &= \min\Big\{\delta_1''(v_{i_j,t}) + \sum_{k=0}^{t-1} h_{i_j,k} \ \Big|\ v_{i_j,t} \in \mathrm{WCA}_{i_j}(\mathfrak{p})\Big\} \\ &\ge \min\Big\{\delta_2''(v_{i_j,t}) + \sum_{k=0}^{t-1} h_{i_j,k} \ \Big|\ v_{i_j,t} \in \mathrm{WCA}_{i_j}(\mathfrak{P})\Big\} = \gamma_{(V_2'',\delta_2'')}(v_i). \end{aligned}$$

So $X_i^{\gamma_{(V_2'',\delta_2'')}(v_i)} \mid X_i^{\gamma_{(V_1'',\delta_1'')}(v_i)}$. Hence $P\big(q(V_1''), \gamma_{(V_1'',\delta_1'')}\big) \subseteq P\big(q(V_2''), \gamma_{(V_2'',\delta_2'')}\big)$. $\square$

Example 3.22. Consider the following two pairs of sets $\mathfrak{p} := (V_1'', \delta_1'') := \{v_{1,1}^4, v_2^5, v_{2,1}^6\}$ and $\mathfrak{P} := (V_2'', \delta_2'') := \{v_{1,1}^3, v_{1,2}^6, v_2^5, v_{2,1}^3\}$ of $(\Sigma_2 P_1)_\lambda$ as in Example 3.16.



Since $V_1'' \subseteq V_2''$ and $\delta_1'' \geq \delta_2''|_{V_1''}$, we have $(V_1'', \delta_1'') \leq (V_2'', \delta_2'')$. Similar to Example 3.16, we have $\mathrm{WCA}_1(\mathfrak{p}) = \{v_{1,1}\}$ and $\mathrm{WCA}_2(\mathfrak{p}) = \emptyset$. So $\gamma_{(V_1'', \delta_1'')}(v_2) = \infty$ and

$$\gamma_{(V_1'', \delta_1'')}(v_1) = \delta_1''(v_{1,1}) + \sum_{k=0}^{1-1} h_{1,k} = \delta_1''(v_{1,1}) + \max\{\lambda(v_1 v_2), \lambda(v_1 v_{1,1})\} = 4 + \max\{2, 2\} = 5.$$

Also, since $q(V_1'') = \{v_1, v_2\}$, we have $P\big(q(V_1''), \gamma_{(V_1'', \delta_1'')}\big) = (X_1^5, X_2^\infty)R = (X_1^5)R$. Then from Example 3.16 we have

$$P\big(q(V_2''), \gamma_{(V_2'', \delta_2'')}\big) = (X_1^5, X_2^6)R \supseteq (X_1^5)R = P\big(q(V_1''), \gamma_{(V_1'', \delta_1'')}\big).$$

Here is our second decomposition result for computing $\mathrm{type}\big(R'/I_r((\Sigma_r G)_\lambda)\big)$.

Corollary 3.23. Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω such that $\lambda(v_i v_j) \leq \lambda(v_i, v_{i,1})$ and $\lambda(v_i v_j) \leq \lambda(v_j, v_{j,1})$ for all edges $v_i v_j \in E$. One has

$$I_r\big((\Sigma_{r-1} G)_{\lambda'}\big)R = \bigcap_{(V'', \delta'') \text{ min. w. } r\text{-path v. cover of } (\Sigma_{r-1} G)_{\lambda'}} P\big(q(V''), \gamma_{(V'', \delta'')}\big), \text{ where } \lambda' = \lambda|_{\Sigma_{r-1} G},$$

and

$$I_r\big((\Sigma_r G)_\lambda\big)R = \bigcap_{(V'', \delta'') \text{ min. w. } r\text{-path v. cover of } (\Sigma_{r-1} G)_{\lambda'}} P\big(q(V''), \gamma_{(V'', \delta'')}\big) + \mathfrak{m}^{[\underline{a}(\lambda)]}.$$

Proof. By Fact 3.11 and [5, Theorem 7.5.3], it is enough to prove that

$$I_r\big((\Sigma_{r-1} G)_{\lambda'}\big)R = \bigcap_{(V'', \delta'') \text{ min. w. } r\text{-path v. cover of } (\Sigma_{r-1} G)_{\lambda'}} P\big(q(V''), \gamma_{(V'', \delta'')}\big).$$

By Theorem 3.19, it is enough to show that

$$\begin{aligned} &\bigcap_{(V'', \delta'') \text{ weighted. } r\text{-path v. cover of } (\Sigma_{r-1} G)_{\lambda'}} P\big(q(V''), \gamma_{(V'', \delta'')}\big) \\ = &\bigcap_{(V'', \delta'') \text{ min. weighted } r\text{-path v. cover of } (\Sigma_{r-1} G)_{\lambda'}} P\big(q(V''), \gamma_{(V'', \delta'')}\big). \end{aligned}$$

"$\subseteq$" follows because every minimal weighted r-path vertex cover is a weighted r-path vertex cover.

"$\supseteq$" follows from [4, Lemma 1.11] and Lemma 3.21. $\square$

Example 3.24. Consider the following weighted 3-path suspension $(\Sigma_3 P_1)_\lambda$ of the weighted 1-path $(P_1)_\omega = \big(\ v_1 \overset{2}{\text{———}} v_2\ \big)$.

$$\begin{array}{ccccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2} & \overset{2}{\text{———}} & v_{1,3} \\ {\scriptstyle 2}\big| & & & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} & \overset{2}{\text{———}} & v_{2,3} \end{array}$$

We depict the minimal weighted 3-path vertex covers of $(\Sigma_2 P_1)_{\lambda'}$ with $\lambda' = \lambda|_{\Sigma_2 P_1}$ in the following sketches.

$$\begin{array}{ccccc} v_1^2 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\big| & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} \end{array} \qquad \begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\big| & & & & \\ v_2^2 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} \end{array} \qquad \begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1}^5 & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\big| & & & & \\ v_2^3 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} \end{array}$$

$$\begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2}^5 \\ {\scriptstyle 2}\big| & & & & \\ v_2^3 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} \end{array} \qquad \begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1}^2 & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\big| & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1}^4 & \overset{4}{\text{———}} & v_{2,2} \end{array} \qquad \begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1}^5 & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\big| & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1}^3 & \overset{4}{\text{———}} & v_{2,2} \end{array}$$

$$\begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2}^5 \\ {\scriptstyle 2}\big| & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1}^3 & \overset{4}{\text{———}} & v_{2,2} \end{array} \qquad \begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1}^2 & \overset{5}{\text{———}} & v_{1,2} \\ {\scriptstyle 2}\big| & & & & \\ v_2^2 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2}^4 \end{array}$$

Since $I_3\big((\Sigma_2 P_1)_{\lambda'}\big) = (X_{1,2}^5 X_{1,1}^5 X_1^2 X_2^2, X_{1,1}^2 X_1^2 X_2^3 X_{2,1}^3, X_1^2 X_2^3 X_{2,1}^4 X_{2,2}^4)$, by Corollary 3.23, we have

$$\begin{aligned} I_3\big((\Sigma_2 P_1)_{\lambda'}\big)R &= \big(X_1^{12}X_2^2, X_1^4X_2^6, X_1^2X_2^{11}\big)R = \big(X_1^2\big)R \cap \big(X_2^2\big)R \cap \big(X_1^7, X_2^3\big)R \cap \big(X_1^{12}, X_2^3\big)R \\ &\quad \cap \big(X_1^4, X_2^7\big) \cap \big(X_1^7, X_2^6\big) \cap \big(X_1^{12}, X_2^6\big)R \cap \big(X_1^4, X_2^{11}\big)R. \end{aligned}$$

Thus, the decomposition in Corollary 3.23 may be redundant.

In light of the preceding example, we define another order from which we can produce an irredundant decomposition. Lemma 3.33 is the key for understanding how this ordering helps with irredundancy.

Definition 3.25. Given two minimal weighted r-path vertex covers $(V_1'', \delta_1''), (V_2'', \delta_2'')$ of $(\Sigma_{r-1}G)_\lambda$, we write $(V_1'', \delta_1'') \leq_p (V_2'', \delta_2'')$ if $q(V_1'') \subseteq q(V_2'')$ and $\gamma_{(V_1'', \delta_1'')} \geq \gamma_{(V_2'', \delta_2'')}|_{q(V_1'')}$. A minimal weighted r-path vertex cover (V'', δ'') is *p-minimal* if there is not another minimal weighted r-path vertex cover (V''', δ''') such that $(V'', \delta'') <_p (V''', \delta''')$.

Example 3.26. Consider the following two minimal weighted 3-path vertex covers $(V_1'', \delta_1'') := \{v_{1,1}^5, v_2^3\}$ and $(V_2'', \delta_2'') := \{v_{1,2}^5, v_2^3\}$ of $(\Sigma_2 P_1)_\lambda$ as in Example 3.16.

$$\begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2}^5 \\ {\scriptstyle 2}\big| & & & & \\ v_2^3 & \overset{3}{\text{———}} & v_{2,1} & \overset{4}{\text{———}} & v_{2,2} \end{array} \qquad \begin{array}{ccccc} v_1 & \overset{2}{\text{———}} & v_{1,1} & \overset{5}{\text{———}} & v_{1,2}^5 \\ {\scriptstyle 2}\big| & & & & \\ v_2 & \overset{3}{\text{———}} & v_{2,1}^3 & \overset{4}{\text{———}} & v_{2,2} \end{array}$$

Then $q(V_1'') = \{v_1, v_2\} = q(V_2'')$. Since

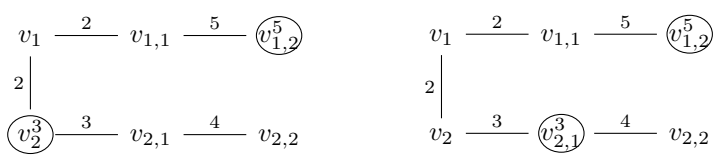
$$\gamma_{(V_1'',\delta_1'')}(v_1) = \delta_1''(v_{1,2}) + h_{1,1} + h_{1,0} = 5 + 5 + 2 = \delta_2''(v_{1,2}) + h_{1,1} + h_{1,0} = \gamma_{(V_2'',\delta_2'')}(v_1),$$

and $\gamma_{(V_1'',\delta_1'')}(v_2) = \delta_1''(v_2) = 3 < 3+3 = \delta_2''(v_{2,1}) + h_{2,0} = \gamma_{(V_2'',\delta_2'')}(v_2)$, we have $\gamma_{(V_1'',\delta_1'')} < \gamma_{(V_2'',\delta_2'')}$. So $(V_1'', \delta_1'') >_{\mathscr{p}} (V_2'', \delta_2'')$. Hence (V_1'', δ_1'') is not $\mathscr{p}$-minimal.

Lemma 3.27. Let $\mathfrak{p} := (W', \delta'), \mathfrak{P} := (W'', \delta'')$ be two minimal weighted r-path vertex covers of $(\Sigma_{r-1}G)_\lambda$ such that $(W'', \delta'') \leq_{\mathscr{p}} (W', \delta')$, then $|(W'', \delta'')| = |(W', \delta')|$ and $q(W'') = q(W')$.

Proof. Since W' is a minimal r-path vertex cover of $\Sigma_{r-1}G$, for distinct $v_{i_1,j_1}, v_{i_2,j_2} \in W'$, we have $i_1 \neq i_2$. Also, since $q(W'') \subseteq q(W')$, $|W''| = |q(W'')| \leq |q(W')| = |W'|$. Suppose $|W''| < |W'|$. Then there exists $v_{i,j} \in W'$ such that $v_i \notin q(W'')$. Since W' is a minimal r-path vertex cover of $\Sigma_{r-1}G$, there is an r-path P_r in $\Sigma_{r-1}G$ that can only be covered by $v_{i,j}$. By assumption, P_r can be covered by some $v_{k,l} \in W''$, so $v_k \in q(W'')$. Also, since $v_i \notin q(W'')$, we have $k \neq i$. Since $\gamma_{W'}(v_k) \leq \gamma_{W''}(v_k)$, we have $v_{k,t} \in W'$ for some $t = \gamma_{W'}(v_k) \leq \gamma_{W''}(v_k) = l$. Note that P_r can also be covered by $v_{k,t} \in W'$, a contradiction. Hence $|W''| = |W'|$ and thus $|q(W'')| = |q(W')|$. Since $q(W'') \subseteq q(W')$, we have $q(W'') = q(W')$. □

Example 3.28. Consider the following two minimal weighted 3-path vertex covers $(V_1'', \delta_1'') := \{v_{1,1}^5, v_2^3\}$ and $(V_2'', \delta_2'') := \{v_{1,2}^5, v_2^3\}$ of $(\Sigma_2 P_1)_\lambda$ as in Example 3.30(a).



By Example 3.30(a), $(V_1'', \delta_1'') <_{\mathscr{p}} (V_2'', \delta_2'')$. Then $|(V_1'', \delta_1'')| = |\{v_{1,2}, v_2\}| = 2 = |\{v_{1,2}, v_{2,1}\}| = |(V_2'', \delta_2'')|$ and $q(V_1'') = \{v_1, v_2\} = q(V_2'')$.

The following theorem can be used as an algorithm to find the set of $\mathscr{p}$-minimal weighted r-path vertex covers of $(\Sigma_{r-1}G)_\lambda$ from the set of minimal weighted r-path vertex covers.

Theorem 3.29. *Let $\mathfrak{p} := (V_1'', \delta_1''), \mathfrak{P} := (V_2'', \delta_2'')$ be two minimal weighted r-path vertex covers of $(\Sigma_{r-1}G)_\lambda$. Then $(V_1'', \delta_1'') \leq_{\mathscr{p}} (V_2'', \delta_2'')$ if and only if $q(V_1'') = q(V_2'')$ and for any $v_{i_l} \in q(V_1'')$: $j_{1,l} > j_{2,l}$ or $j_{1,l} = j_{2,l}$ and $\delta_1''(v_{i_l,j_{1,l}}) \geq \delta_2''(v_{i_l,j_{2,l}})$ with $j_{1,l} := \{j \mid v_{i_l,j} \in V_1''\}$ and $j_{2,l} = \{j \mid v_{i_l,j} \in V_2''\}$.*

Proof. By Lemma 3.27, $(V_1'', \delta_1'') \leq_{\mathscr{p}} (V_2'', \delta_2'')$ if and only if $q(V_1'') = q(V_2'')$ and $\gamma_{(V_1'',\delta_1'')}|_{q(V_1'')} \geq \gamma_{(V_2'',\delta_2'')}|_{q(V_1'')}$ if and only if $q(V_1'') = q(V_2'')$ and for any $v_{i_l} \in q(V_1'')$, $\gamma_{(V_1'',\delta_1'')}(v_{i_l}) \geq \gamma_{(V_2'',\delta_2'')}(v_{i_l})$ if and only if $q(V_1'') = q(V_2'')$ and for any $v_{i_l} \in q(V_1'')$, $\delta_1''(v_{i_l,j_{1,l}}) + \sum_{k=0}^{j_{1,l}-1} h_{i_l,k} \geq \delta_2''(v_{i_l,j_{2,l}}) + \sum_{k=0}^{j_{2,l}-1} h_{i_l,k}$ by Proposition 3.14. Claim. For $v_{i_l} \in q(V_1'')$, $\delta_1''(v_{i_l,j_{1,l}}) + \sum_{k=0}^{j_{1,l}-1} h_{i_l,k} \geq \delta_2''(v_{i_l,j_{2,l}}) + \sum_{k=0}^{j_{2,l}-1} h_{i_l,k}$ if and only if $j_{1,l} > j_{2,l}$, or $j_{1,l} = j_{2,l}$ and $\delta_1''(v_{i_l,j_{1,l}}) \geq \delta_2''(v_{i_l,j_{2,l}})$. Then we are done.

"$\Leftarrow$". Assume $j_{1,l} > j_{2,l}$, or $j_{1,l} = j_{2,l}$ and $\delta_1''(v_{i_l,j_{1,l}}) \geq \delta_2''(v_{i_l,j_{2,l}})$. Then

$$\alpha := \left(\delta_1''(v_{i_l,j_{1,l}}) + \sum_{k=0}^{j_{1,l}-1} h_{i_l,k}\right) - \left(\delta_2''(v_{i_l,j_{2,l}}) + \sum_{k=0}^{j_{2,l}-1} h_{i_l,k}\right) = \delta_1''(v_{i_l,j_{1,l}}) - \delta_2''(v_{i_l,j_{2,l}}) + \sum_{k=j_{2,l}}^{j_{1,l}-1} h_{i_l,k}.$$

To prove our statement, it is equivalent to show $\alpha \geq 0$.

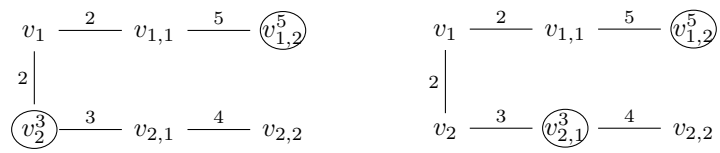
(a) If $j_{1,l} > j_{2,l}$, then $\alpha \geq \delta_1''(v_{i_l,j_{1,l}}) - \delta_2''(v_{i_l,j_{2,l}}) + h_{i_l,j_{2,l}} > h_{i_l,j_{2,l}} - \delta_2''(v_{i_l,j_{2,l}}) \geq 0$.
(b) If $j_{1,l} = j_{2,l}$ and $\delta_1''(v_{i_l,j_{1,l}}) \geq \delta_2''(v_{i_l,j_{2,l}})$, then $\alpha = \delta_1''(v_{i_l,j_{1,l}}) - \delta_2''(v_{i_l,j_{2,l}}) \geq 0$.

"$\Rightarrow$". Suppose $j_{1,l} < j_{2,l}$, or $j_{1,l} = j_{2,l}$ and $\delta_1''(v_{i_l,j_{1,l}}) < \delta_2''(v_{i_l,j_{2,l}})$. Then

$$\alpha := \left(\delta_1''(v_{i_l,j_{1,l}}) + \sum_{k=0}^{j_{1,l}-1} h_{i_l,k}\right) - \left(\delta_2''(v_{i_l,j_{2,l}}) + \sum_{k=0}^{j_{2,l}-1} h_{i_l,k}\right) = \delta_1''(v_{i_l,j_{1,l}}) - \delta_2''(v_{i_l,j_{2,l}}) - \sum_{k=j_{1,l}}^{j_{2,l}-1} h_{i_l,k}.$$

To prove our statement, it is equivalent to show $\alpha < 0$.
(a) If $j_{1,l} = j_{2,l}$ and $\delta_1''(v_{i_l,j_{1,l}}) < \delta_2''(v_{i_l,j_{2,l}})$, then $\alpha = \delta_1''(v_{i_l,j_{1,l}}) - \delta_2''(v_{i_l,j_{2,l}}) < 0$.
(b) Assume $j_{1,l} < j_{2,l}$. Since $v_{i_l,j_{1,l}} \in V_1''$ and V_1'' is a minimal weighted r-path vertex cover, $\delta_1''(v_{i_l}, j_{1,l}) \leq h_{i_l,j_{1,l}}$. So $\alpha = \delta_1''(v_{i_l,j_{1,l}}) - \delta_2''(v_{i_l,j_{2,l}}) - \sum_{k=j_{1,l}}^{j_{2,l}-1} h_{i_l,k} < \delta_1''(v_{i_l,j_{1,l}}) - h_{i_l,j_{1,l}} \leq 0$. $\square$

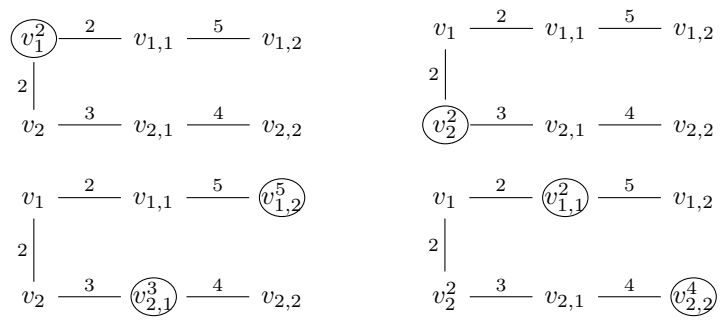
Example 3.30. We have the following examples.
(a) Consider the following two minimal weighted 3-path vertex covers $(V_1'', \delta_1'') := \{v_{1,1}^5, v_2^3\}$ and $(V_2'', \delta_2'') := \{v_{1,2}^5, v_2^3\}$ of $(\Sigma_2 P_1)_\lambda$ as in Example 3.26.



Then $q(V_1'') = \{v_{i_1} := v_1, v_{i_2} := v_2\} = q(V_2'')$. Note that

$$j_{1,1} = \min\{j \mid v_{i_1,j} \in V_1''\} = \min\{j \mid v_{1,j} \in V_1''\} = 2,$$
$$j_{1,2} = \min\{j \mid v_{i_2,j} \in V_1''\} = \min\{j \mid v_{2,j} \in V_1''\} = 0,$$
$$j_{2,1} = \min\{j \mid v_{i_1,j} \in V_2''\} = \min\{j \mid v_{2,j} \in V_2''\} = 2,$$
$$j_{2,2} = \min\{j \mid v_{i_2,j} \in V_2''\} = \min\{j \mid v_{2,j} \in V_2''\} = 1.$$

Since $j_{1,1} = 2 = j_{2,1}$ and $\delta_1''(v_{1,j_{1,1}}) = \delta_1''(v_{1,2}) = 5 = \delta_2''(v_{1,2}) = \delta_2''(v_{1,j_{2,1}})$, and $j_{1,2} = 0 < 1 = j_{2,2}$, we have $(V_1'', \delta_1'') <_{\mathscr{p}} (V_2'', \delta_2'')$ by Theorem 3.29.
(b) Consider all the minimal weighted 3-path vertex covers of $(\Sigma_2 P_1)_{\lambda'}$ as in Example 3.24. Apply Theorem 3.29 repeatedly, we get all the $\mathscr{p}$-minimal weighted 3-path vertex covers in the following.

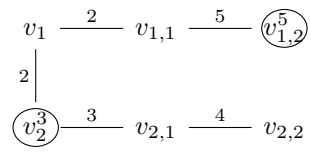


The next two results are key for our third and final decomposition result.

Proposition 3.31. For every minimal weighted r-path vertex cover $\mathfrak{p} := (W', \delta')$ of $(\Sigma_{r-1}G)_\lambda$, there is a $\mathscr{p}$-minimal weighted r-path vertex cover (W'', δ'') of $(\Sigma_{r-1}G)_\lambda$ such that $(W'', \delta'') \leq_{\mathscr{p}} (W', \delta')$.

Proof. If (W', δ') is itself a $\mathscr{p}$-minimal weighted r-path vertex cover for $(\Sigma_{r-1}G)_\lambda$, then we are done. If (W', δ') is not $\mathscr{p}$-minimal, then by Lemma 3.27, for some $v_i \in q(W')$ the function $\gamma_{(W',\delta')}(v_i) = \delta'(v_{i,j_0}) + \sum_{k=0}^{j_0-1} h_{i,k}$ with $j_0 := \{j \mid v_{i,j} \in \mathrm{WCA}_i(\mathfrak{p})\}$ from Proposition 3.14 can be increased, which is done by increasing j_0 and assigning an appropriate value to $\delta'(v_{i,j_0})$ since (W', δ') is minimal. We increase $\gamma_{(W',\delta')}(v_i)$ for each $v_i \in q(W')$ such that any further increase would cause the set not to be a weighted r-path vertex cover. This process terminates in finitely many steps because $j_0 \leq r$. Denote the new set (W'', δ''). Then (W'', δ'') is minimal since the size of W'' cannot be decreased by Lemma 3.27 and δ'' cannot be increased. Thus, by construction, (W'', δ'') is a $\mathscr{p}$-minimal weighted r-path vertex cover for $(\Sigma_{r-1}G)_\lambda$ such that $(W'', \delta'') \leq_{\mathscr{p}} (W', \delta')$. □

Example 3.32. Consider the following minimal weighted 3-path vertex cover $\mathfrak{p} := (V_1'', \delta_1'') := \{v_{1,1}^5, v_2^3\}$ of $(\Sigma_2 P_1)_\lambda$ as in Example 3.30(a).

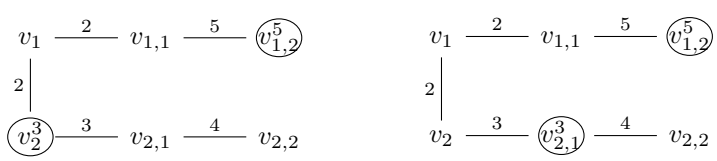


Note that $\gamma_{(V_1'', \delta_1'')}(v_1)$ cannot be increased. Assume $v_{2,1} \in V''$. Then set $\delta''(v_{2,1}) = 3$, we have $\mathfrak{p}' := (V_1''', \delta_1''') = \{v_{1,2}^5, v_{2,1}^3\}$ is a minimal weighted 3-path vertex cover by Example 3.30(a). However, since $v_{1,2} \in V''$, we have $v_{2,2}$ cannot be in V''', otherwise the 3-path $v_{1,1}v_1v_2v_{2,1}$ will be left uncovered. Thus, (V_1''', δ_1''') is $\mathscr{p}$-minimal and $(V_1''', \delta_1''') <_{\mathscr{p}} (V_1'', \delta_1'')$.

Lemma 3.33. Let $(V_1', \delta_1'), (V_2', \delta_2')$ be two minimal weighted r-path vertex covers of $(\Sigma_{r-1}G)_\lambda$. Then $(V_1', \delta_1') \leq_{\mathscr{p}} (V_2', \delta_2')$ if and only if $P\big(q(V_1'), \gamma_{(V_1', \delta_1')}\big) \subseteq P\big(q(V_2'), \gamma_{(V_2', \delta_2')}\big)$.

Proof. $(V_1', \delta_1') \leq_{\mathscr{p}} (V_2', \delta_2')$ if and only if $q(V_1') \subseteq q(V_2')$ and $\gamma_{(V_1', \delta_1')}|_{q(V_1')} \geq \gamma_{(V_2', \delta_2')}|_{q(V_1')}$ if and only if $P\big(q(V_1'), \gamma_{(V_1', \delta_1')}\big) \subseteq P\big(q(V_2'), \gamma_{(V_2', \delta_2')}\big)$. □

Example 3.34. Consider the following two minimal weighted 3-path vertex covers $(V_1'', \delta_1'') := \{v_{1,1}^5, v_2^3\}$ and $(V_2'', \delta_2'') := \{v_{1,2}^5, v_2^3\}$ of $(\Sigma_2 P_1)_\lambda$ as in Example 3.30(a).



Then $(V_2'', \delta_2'') <_{\mathscr{p}} (V_1'', \delta_1'')$ by Example 3.30(a). Also note that

$$P\big(q(V_2''), \gamma_{(V_2'', \delta_2'')}\big) = \big(X_1^{12}, X_2^6\big)R \subseteq \big(X_1^{12}, X_2^3\big)R = P\big(q(V_1''), \gamma_{(V_1'', \delta_1'')}\big).$$

Next, we present our third and final decomposition result which will yield the type computation in Theorem 3.37.

Theorem 3.35. *Let $(\Sigma_r G)_\lambda$ be a weighted r-path suspension of G_ω such that $\lambda(v_iv_j) \leq \lambda(v_i, v_{i,1})$ and $\lambda(v_iv_j) \leq \lambda(v_j, v_{j,1})$ for all edges $v_iv_j \in E$. One has an irredundant parametric decomposition*

$$I_r\big((\Sigma_r G)_\lambda\big)R = \bigcap_{(V'', \delta'') \ \mathscr{p}\text{-min. w. } r\text{-path v. c. of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''), \gamma_{(V'', \delta'')}\big) + \mathfrak{m}^{[\underline{a}(\lambda)]}, \ \lambda' = \lambda|_{\Sigma_{r-1}G}.$$

Proof. By Fact 3.11 and [5, Theorem 7.5.3], to verify this result, it is enough to show that we have an irredundant decomposition

$$I_r\big((\Sigma_{r-1}G)_{\lambda'}\big)R = \bigcap_{(V'',\delta'')\ \mathscr{p}\text{-min. w. } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''),\gamma_{(V'',\delta'')}\big).$$

Lemma 3.33 shows that this intersection is irredundant. So by Corollary 3.23, it is enough to show that

$$\begin{aligned}&\bigcap_{(V'',\delta'')\text{ min. weighted } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''),\gamma_{(V'',\delta'')}\big)\\ =&\bigcap_{(V'',\delta'')\ \mathscr{p}\text{-min. weighted } r\text{-path v. cover of } (\Sigma_{r-1}G)_{\lambda'}} P\big(q(V''),\gamma_{(V'',\delta'')}\big).\end{aligned}$$

"$\subseteq$" follows as every $\mathscr{p}$-minimal weighted r-path vertex cover is a minimal weighted r-path vertex cover.

"$\supseteq$" follows from Proposition 3.31 and Lemma 3.33. □

Example 3.36. Consider the graph $(\Sigma_3 P_1)_\lambda$ as in Example 3.24. Then by Theorem 3.35 and Example 3.30(b), we have an irredundant parametric decomposition

$$\begin{aligned}I_3\big((\Sigma_3P_1)_\lambda\big) &= \big(X_1^{12}X_2^2, X_1^4X_2^6, X_1^2X_2^{11}\big)R + \mathfrak{m}^{[\underline{a}(\lambda)]}\\ &= \big[(X_1^2)R\cap(X_2^2)R\cap(X_1^{12},X_2^6)R\cap(X_1^4,X_2^{11})R\big] + (X_1^{14},X_2^{13})R.\end{aligned}$$

The next theorem is the first main result of this paper. It is Formula (*) from the abstract.

Theorem 3.37. *Let $(\Sigma_rG)_\lambda$ be a weighted r-path suspension of G_ω such that $\lambda(v_iv_j)\le\lambda(v_i,v_{i,1})$ and $\lambda(v_iv_j)\le\lambda(v_j,v_{j,1})$ for all edges $v_iv_j\in E$.*

$$\operatorname{type}\left(\frac{R'}{I_r((\Sigma_rG)_\lambda)}\right) = \sharp\,\{\mathscr{p}\text{-minimal weighted } r\text{-path vertex covers of } (\Sigma_{r-1}G)_{\lambda'}\},\ \ \lambda'=\lambda|_{\Sigma_{r-1}G}.$$

Proof. We compute

$$\begin{aligned}\operatorname{type}\left(\frac{R'}{I_r((\Sigma_rG)_\lambda)}\right) &= \operatorname{type}\left(\frac{R'}{I_r((\Sigma_rG)_\lambda)+(X_i-X_{i,k}\mid 1\le i\le d, 1\le k\le r)R'}\right)\\ &= \operatorname{type}\left(\frac{R}{I_r((\Sigma_rG)_\lambda)R}\right)\\ &= \sharp\,\{\text{ideals in an irredundant parametric decomposition of } I_r\big((\Sigma_rG)_\lambda\big)R\}\\ &= \sharp\,\{\mathscr{p}\text{-minimal weighted } r\text{-path vertex covers of } (\Sigma_{r-1}G)_{\lambda'}\},\end{aligned}$$

where the first equality is from [3, Proposition A.6.2], 3.12(a) and 3.11, the second equality is from Fact 3.11, the third equality is from Fact 2.2 since $\dim\big(\frac{R}{I_r((\Sigma_rG)_\lambda)R}\big)=0$, and the last equality is from Fact 3.35. □

Remark. Because of Fact 3.12, we can use Theorem 3.37 to compute $\operatorname{type}\big(R/I_r(G_\omega)\big)$ for all weighted trees G_ω such that $I_r(G_\omega)$ is Cohen-Macaulay.

Example 3.38. Consider Example 3.36. Then by Theorem 3.37, we have

$$\operatorname{type}\big(R'/I_3(\Sigma_3P_1)_\lambda\big)=4.$$

Observe that the smallest number of vertices for one of the 3-path vertex covers of $(\Sigma_3 P_1)_\lambda$ is 2. Then by [4, Theorem 2.7] and [5, Theorem 5.1.2], $\dim\big(R'/I_3((\Sigma_3 P_1)_\lambda)\big) = 8-2 = 6$. Since $R'/I_3((\Sigma_3 P_1)_\lambda)$ is Cohen-Macaulay by Fact 3.12(a), $\operatorname{depth}\big(R'/I_3((\Sigma_3 P_1)_\lambda)\big) = \dim\big(R'/I_3((\Sigma_3 P_1)_\lambda)\big) = 6$. Hence

$$\operatorname{Ext}^6_{R'}\big(\mathbb{K}, R'/I_3((\Sigma_3 P_1)_\lambda)\big) \cong \mathbb{K}^4.$$

Corollary 3.39. We have the following.

(a)

$$\operatorname{type}\left(\frac{R'}{I_r(\Sigma_r G)}\right) = \sharp\,\{\mathscr{p}\text{-minimal } r\text{-path vertex covers of } \Sigma_{r-1}G\}.$$

(b) Let $(\Sigma G)_\lambda$ be a weighted suspension of G_ω such that $\lambda(v_iv_j) \le \lambda(v_iw_i)$ and $\lambda(v_iv_j) \le \lambda(w_jv_j)$ for each $v_iv_j \in E$. Then

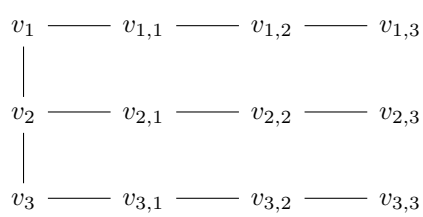
$$\operatorname{type}\left(\frac{R'}{I((\Sigma G)_\lambda)}\right) = \sharp\,\{\text{minimal weighted vertex covers of } G_\omega\}.$$

(c)

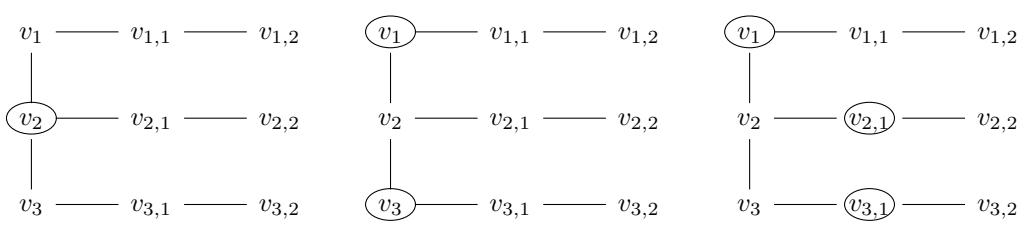
$$\operatorname{type}\left(\frac{R'}{I(\Sigma G)}\right) = \sharp\,\{\text{minimal vertex covers of } G\}.$$

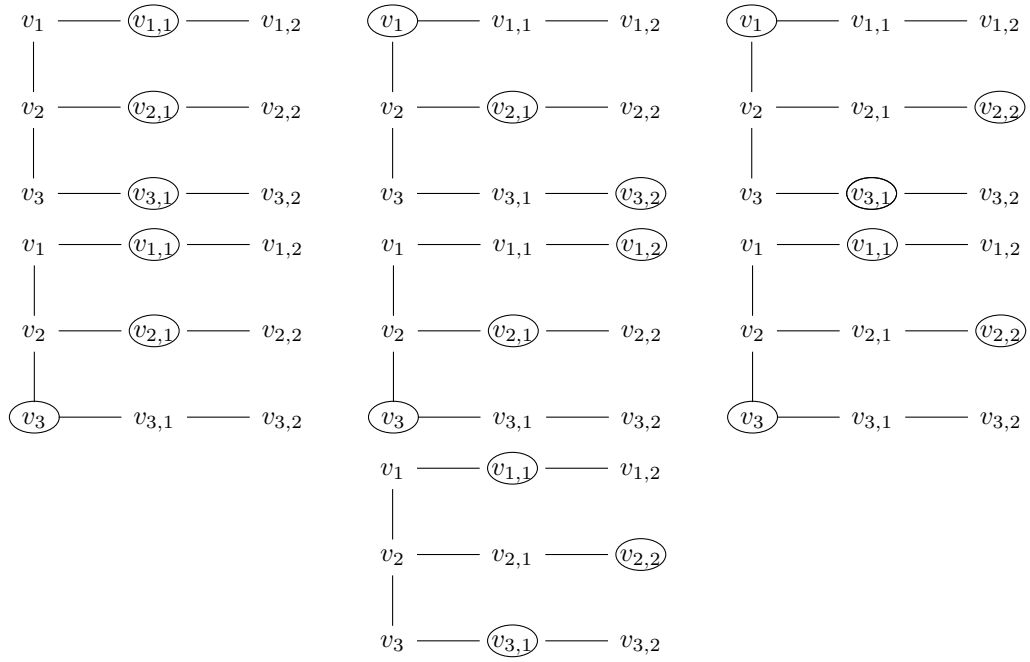
Proof. (a) Let $1 : E(\Sigma_r G) \to \mathbb{N}$ be the constant weight function on $\Sigma_r G$ defined by $1(e) = 1$ for $e \in E(\Sigma_r G)$. Then $\Sigma_{r-1}G = (\Sigma_{r-1}G)_1$ and $I_r(\Sigma_r G) = I_r((\Sigma_r G)_1)$. So the conclusion is covered in Theorem 3.37.
(b) Let (V'', δ'') be a minimal 1-path vertex cover of G_ω. By definition, any $\mathscr{p}$-minimal 1-path vertex cover of G_ω is a minimal 1-path vertex cover. So it suffices to show that (V'', δ'') is a $\mathscr{p}$-minimal 1-path vertex cover of G_ω, which is true by Theorem 3.29.
(c) Let $1 : E(\Sigma G) \to \mathbb{N}$ be the constant weight function on ΣG defined by $1(e) = 1$ for $e \in E(\Sigma G)$. Then $G = G_1$ and $I(\Sigma G) = I((\Sigma G)_1)$. So the conclusion is covered in (b). □

Example 3.40. (a) Consider the following graph $\Sigma_3 P_2$ with $P_2 = \big(\ v_1 \text{ —— } v_2 \text{ —— } v_3\ \big)$



We depict the minimal 3-path vertex covers of $\Sigma_2 P_2$ in the following sketches.

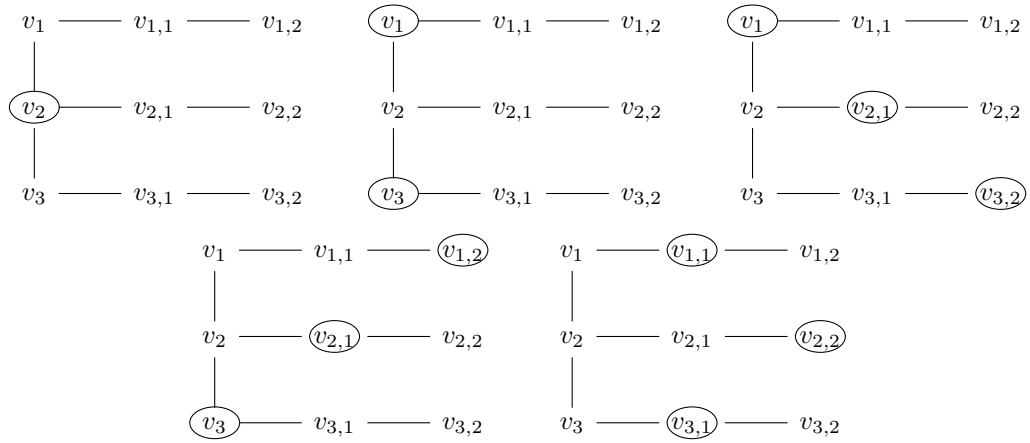




So by Corollary 3.23, we have a parametric decomposition

$$\begin{aligned} I_3(\Sigma_3 P_2)R = {} & (X_2)R \cap (X_1, X_3)R \cap (X_1, X_2^2, X_3^2)R \cap (X_1^2, X_2^2, X_3^2)R \cap (X_1, X_2^2, X_3^3)R \\ & \cap (X_1, X_2^3, X_3^2)R \cap (X_1^2, X_2^2, X_3)R \cap (X_1^3, X_2^2, X_3)R \cap (X_1^2, X_2^3, X_3)R \\ & \cap (X_1^2, X_2^3, X_3^2)R + \mathfrak{m}^3, \end{aligned}$$

which is a redundant decomposition since e.g., the last ideal $(X_1^2, X_2^3, X_3^2)R$ is contained in the second to last ideal $(X_1^2, X_2^3, X_3)R$. Note that the $\mathscr{p}$-minimal 3-path vertex covers of $\Sigma_2 P_2$ are the following.



So by Theorem 3.35 and we have an irredundant parametric decomposition

$$I_3(\Sigma_3 P_2)R = \left[(X_2)R \cap (X_1, X_3)R \cap (X_1, X_2^2, X_3^3)R \cap (X_1^3, X_2^2, X_3)R \cap (X_1^2, X_2^3, X_3^2)R\right] + \mathfrak{m}^3,$$

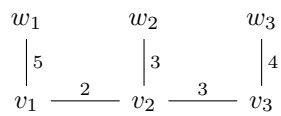
and by Corollary 3.39(a), we have

$$\operatorname{type}\big(R'/I_3(\Sigma_3 P_2)\big) = 5.$$

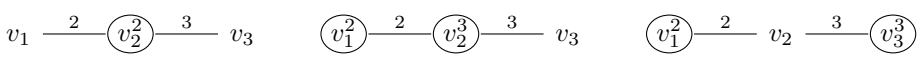
We observe that the smallest number of vertices for one of the 3-path vertex covers of $\Sigma_3 P_2$ is 3. Then by [4, Theorem 2.7] and [5, Theorem 5.1.2], $\dim\big(R'/I_3(\Sigma_3 P_2)\big) = 12 - 3 = 9$. Since $R'/I_3(\Sigma_3 P_2)$ is Cohen-Macaulay by Fact 3.12(a), $\operatorname{depth}(R'/I_3(\Sigma_3 P_2)) = \dim\big(R'/I_3(\Sigma_3 P_2)\big) = 9$. Hence

$$\operatorname{Ext}^9_{R'}\big(\mathbb{K}, R'/I_3(\Sigma_3 P_2)\big) \cong \mathbb{K}^5.$$

(b) Consider the following weighted graph $(\Sigma P_2)_\lambda$ with $(P_2)_\omega = \big(\ v_1 \overset{2}{\text{———}} v_2 \overset{3}{\text{———}} v_3\ \big)$.



The minimal weighted vertex covers of $(P_2)_\omega = \Big(\ v_1 \overset{2}{\text{———}} v_2 \overset{3}{\text{———}} v_3\ \Big)$ are displayed in the following sketches.



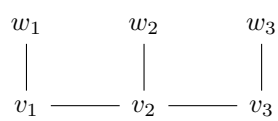
Then by Corollary 3.39(b),

$$\operatorname{type}\big(R'/I((\Sigma P_2)_\lambda)\big) = \sharp\,\{\text{minimal weighted vertex covers of } (P_2)_\omega = 3\}.$$

We observe that the smallest number of vertices for one of the weighted vertex covers of $(\Sigma P_2)_\lambda$ is 3. Then by [4, Theorem 2.7] and [5, Theorem 5.1.2], $\dim\big((R'/I((\Sigma P_2)_\lambda)\big) = 6 - 3 = 3$. Since $R'/I((\Sigma P_2)_\lambda)$ is Cohen-Macaulay by Fact 3.12(a), $\operatorname{depth}(R'/I((\Sigma P_2)_\lambda)) = \dim\big(R'/I((\Sigma P_2)_\lambda)\big) = 3$. Hence

$$\operatorname{Ext}^3_{R'}\big(\mathbb{K}, R'/I((\Sigma P_2)_\lambda)\big) \cong \mathbb{K}^3.$$

(c) Consider the following graph ΣP_2 with $P_2 = \big(\ v_1 \text{———} v_2 \text{———} v_3\ \big)$.



We depict the minimal vertex covers of P_2 in the following sketches.

By Corollary 3.39(c),

$$\operatorname{type}\big(R'/I(\Sigma P_2)\big) = \sharp\,\{\text{minimal vertex covers of } P_2\} = 2.$$

Observe that the smallest number of vertices for one of the vertex covers of ΣP_2 is 3. Then by [4, Theorem 2.7] and [5, Theorem 5.1.2], $\dim\big(R'/I_3((\Sigma_3 P_1)_\lambda)\big) = 6 - 3 = 3$. Since $R'/I(\Sigma P_2)$ is Cohen-Macaulay by Fact 3.12(a), $\operatorname{depth}\big(R'/I(\Sigma P_2)\big) = \dim\big(R'/I(\Sigma P_2)\big) = 3$. Hence

$$\operatorname{Ext}^3_{R'}\big(\mathbb{K}, R'/I(\Sigma P_2)\big) \cong \mathbb{K}^2.$$

4. Edge Ideals $I(D)$ of Oriented Graphs D and the Type of $R'/I(\Sigma D)$

Let $D = (V(D), E(D))$ be an oriented graph with the underlying undirected graph G, i.e., D arises from the undirected G by directing every edge from one of its ends to the other. Let v_iv_j represent an edge in D directed from v_i to v_j. Let $S = A[X_1, \dots, X_d, Y_1, \dots, Y_d]$.

Definition 4.1. A *vertex-weight function* on the oriented graph D is a function $\omega : V \to \mathbb{N}$ that assigns a *weight* to each vertex. A *weighted oriented graph* D_ω is an oriented graph D equipped with a vertex-weight function ω. For simplicity, let $\omega_i = \omega(v_i)$ for $i = 1, \dots, d$.

Assumption. For the remainder of this section, we assume D is a weighted oriented graph with vertex-weight function ω such that the underlying graph is G.

Definition 4.2. The *edge ideal* associated to D is the ideal $I(D) \subseteq R$ that is "generated by the edges of D":

$$I(D) = \Big(X_iX_j^{\omega_j} \mid v_iv_j \in E(D)\Big)R.$$

Example 4.3. The edge ideal of $D = (\ v_1^2 \longleftarrow v_2^3 \longrightarrow v_3^5\ )$ is $I(D) = (X_1^2X_2, X_2X_3^5)R$.

Convention 4.4. The Cohen-Macaulay property of $I(D)$ is independent of the weight we assign to a source or a sink. So when we study this property, we shall always assume $\omega(v_i) = 1$ if v_i is a source or sink.

Definition 4.5. A *vertex cover* of D is a subset of $V' \subseteq V$ such that for each edge $v_iv_j \in E(D)$ we have $v_i \in V'$ or $v_j \in V'$. A vertex cover V' is *minimal* if it does not properly contain another vertex cover of D.

Definition 4.6. Let $v_i \in V$. Define the *out-neighbourhood* of v_i by

$$N_D^+(v_i) = \{v_j \in V \mid v_iv_j \in E(D)\},$$

and define the *in-neighbourhood* of v_i by

$$N_D^-(v_i) = \{v_j \in V \mid v_jv_i \in E(D)\}.$$

Furthermore, the *neighbourhood* of v_i is the set

$$N_D(v_i) = N_D^+(v_i) \sqcup N_D^-(v_i).$$

Definition 4.7. [7, Definition 2.3] Let V' be a vertex cover of D. Define

$$L_1(V') = \big\{v_i \in V' \bigm| N_D^+(v_i) \not\subseteq V'\big\},$$
$$L_2(V') = \big\{v_i \in V' \bigm| N_D^-(v_i) \not\subseteq V'\big\} \smallsetminus L_1(V'),$$

and

$$\begin{aligned} L_3(V') &= V' \smallsetminus \{L_1(V') \sqcup L_2(V')\} = \big\{v_i \in V' \bigm| N_D^+(v_i) \subseteq V' \text{ and } N_D^-(v_i) \subseteq V'\big\} \\ &= \{v_i \in V' \mid N_D(v_i) \subseteq V'\}. \end{aligned}$$

Fact 4.8. [7, Proposition 2.5] Let V' be a vertex cover of D. Then $L_3(V') = \emptyset$ if and only if V' is a minimal vertex cover of D.

Definition 4.9. A vertex cover V' of D is *strong* if for each $v_i \in L_3(V')$ there is $v_jv_i \in E(D)$ such that $v_j \in L_2(V') \cup L_3(V')$ with $\omega_j \geq 2$.

Fact 4.10. If V' is a minimal vertex cover, then V' is strong.

Definition 4.11. [7, Definition 3.4] Let V' be a vertex cover of D. Define

$$I_{V'} = \Big(L_1(V') \sqcup \big\{X_j^{\omega_j} \mid v_j \in L_2(V') \sqcup L_3(V')\big\}\Big)R.$$

Fact 4.12. [7, Theorem 3.11]

$$I(D) = \bigcap_{V' \text{ strong vertex cover}} I_{V'},$$

where the intersection is taken over all strong vertex covers of D. The intersection is irredundant.

Example 4.13. Let $D = (\ v_1^2 \longleftarrow v_2^3 \longrightarrow v_3^5\ )$ as in Example 4.3. Then the strong vertex covers are $\{v_1, v_3\}, \{v_2\}$. Since $I_{\{v_1,v_3\}} = (X_1^2, X_3^5)R$ and $I_{\{v_2\}} = (X_2)R$, by Fact 4.12, we have

$$I(D) = I_{\{v_1,v_3\}} \cap I_{\{v_2\}} = \big(X_1^2, X_3^5\big)R \cap \big(X_2\big)R.$$

Definition 4.14. A *suspension* of D is a weighted oriented graph ΣD with vertex set $V \sqcup \{w_1, \ldots, w_d\}$ such that the underlying graph ΣG is a suspension of G.

Example 4.15. A vertex suspension ΣD of $D = (\ v_1^2 \longleftarrow v_2^3 \longrightarrow v_3^5\ )$ is

$$\begin{array}{ccccc} w_1^1 & & w_2^6 & & w_3^3 \\ \uparrow & & \downarrow & & \uparrow \\ v_1^2 & \longleftarrow & v_2^3 & \longrightarrow & v_3^5. \end{array}$$

Fact 4.16. Let ΣD be a suspension of D such that $\omega_i = \omega(v_i) = 1$ for any edge $v_i w_i \in \Sigma D$. Then by the proof of [2, Theorem 3.1], $I(\Sigma D)$ is the polarization of $I(D) + \big(X_1^{1+\omega_1}, \ldots, X_d^{1+\omega_d}\big)R$. So by [1], the list $X_1 - Y_1, \ldots, X_d - Y_d$ is a maximal regular sequence for $\frac{S}{I(\Sigma D)}$ and

$$\frac{R}{I(D) + \big(X_1^{1+\omega_1}, \ldots, X_d^{1+\omega_d}\big)R} \cong \frac{S}{I(\Sigma D) + (X_1 - Y_1, \ldots, X_d - Y_d)S}.$$

Because of the following fact, the main result of this section gives a formula to compute the type$(R/I(D))$ for all trees such that $R/I(D)$ is Cohen-Macaulay.

Fact 4.17. [2, Theorem 3.1] Let ΣD be a suspension of D such that $\omega(v_i) = 1$ for any edge $v_i w_i \in D$.

(a) Then $S/I(\Sigma D)$ is Cohen-Macaulay.
(b) Let Z be a weighted oriented graph such that the underlying simple graph is Γ. If Γ is a tree and $R/I(\Gamma)$ is Cohen-Macaulay, then $\Gamma = \Sigma H$ for some subtree H.

The following theorem is the second main result of this thesis. It is Formula (**) from the abstract.

Theorem 4.18. *Let ΣD be a suspension of D such that $\omega(v_i) = 1$ for any edge $v_i w_i \in D$. Then*

$$\operatorname{type}\left(\frac{S}{I(\Sigma D)}\right) = \sharp\,\{\textit{strong vertex covers of } D\}.$$

Proof. We compute

$$\begin{aligned}
\operatorname{type}\left(\frac{S}{I(\Sigma D)}\right) &= \operatorname{type}\left(\frac{S}{I(\Sigma D)+(X_1-Y_1,\dots,X_d-Y_d)S}\right)\\
&= \operatorname{type}\left(\frac{R}{I(G_\omega)+(X_1^{1+\omega_1},\dots,X_d^{1+\omega_d})R}\right)\\
&= \sharp\,\{\text{ideals in an irredundant parametric decomposition of}\\
&\quad I(D)+\left(X_1^{1+\omega_1},\dots,X_d^{1+\omega_d}\right)\\
&= \sharp\,\{\text{ideals in an irredundant m-irreducible decomposition of } I(D)\}\\
&= \sharp\,\{\text{strong vertex covers of } D\},
\end{aligned}$$

where the first equality is from [3, Proposition A.6.2], 3.12(a) and 3.11, the second equality is from Fact 4.16, the third equality is from Fact 2.2 since $\dim\left(\frac{R}{I(G_\omega)+\left(X_1^{1+\omega_1},\dots,X_d^{1+\omega_d}\right)R}\right)=0$, the fourth equality is from [5, Exercise 7.5.10], and the last equality is from Fact 4.12. □

Example 4.19. Consider the following weighted oriented graph ΣD as in Example 4.15.

$$\begin{tikzcd}
w_1^1 & w_2^6 \arrow[d] & w_3^3 \\
v_1^2 \arrow[u] & v_2^3 \arrow[l] \arrow[r] & v_3^5 \arrow[u]
\end{tikzcd}$$

Then by Theorem 4.18 and Example 4.13, $\operatorname{type}\big(S/I(\Sigma D)\big)=2$.

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